



STRUCTURED ABSTRACT

From Generic Chatbots to Teacher-Configurable AI Tutors: The Design and Evaluation of Novedu in Secondary Education

CONTEXT

Generative AI and large language models (LLMs) are increasingly used by secondary-school students for learning support. Used without adaptation, general-purpose chatbots such as ChatGPT or Claude function as generic information providers rather than pedagogically aligned tutors and may offer support that is insufficiently tailored to students' needs, prior knowledge, or classroom context (Song et al., 2025). Research on pedagogical alignment further shows that such models tend to give direct answers rather than the scaffolded guidance that supports learning (Sonkar et al., 2024), underlining the need to keep pedagogy central when integrating LLMs into teaching (Hu et al., 2025).

PURPOSE OR GOAL

This work-in-progress paper presents Novedu, a teacher-configurable platform for subject-centred AI tutoring developed at HTBLA Leonding (Austria). Novedu lets teachers create specialised tutors for specific topics, units, or assessments, so that support is aligned with the current focus of instruction, students' expected prior knowledge, and the teacher's pedagogical intentions. The study asks how a school can move from exploratory use of generic AI tools toward an institutionally embedded, teacher-configurable tutoring infrastructure, and what requirements, acceptance, and pedagogical fit such a transition entails.

APPROACH

The project follows a requirements-driven design-and-evaluation approach. In the first phase, teachers and students completed a structured survey eliciting expectations, perceived benefits, concerns, and functional requirements for AI-supported learning. The results informed an initial prototype built on open-source components (Open WebUI and LiteLLM), deployed in a pilot and evaluated for acceptance, perceived usefulness, and pedagogical fit. The team then designed Novedu as a dedicated platform organised around educational workflows: integration with school IT systems, monitoring so teachers can review how their tutors are used, feedback loops for improving tutor definitions, and a modular model in which tutors are assembled from reusable text and prompt modules (e.g., Socratic-tutor behaviour or instructional guidelines). As a computer-science school, HTBLA Leonding develops the platform as a cooperative software-engineering project with students and teachers—both an authentic software-development setting and a practically used learning platform.

ACTUAL OR ANTICIPATED OUTCOMES

By the conference, the evaluation of the open-source prototype phase will be completed, while the custom platform and the broader evaluation remain ongoing. We anticipate reporting the elicited requirements, preliminary findings on the prototype's acceptance, perceived usefulness, and pedagogical fit, and the rationale for the transition toward a teacher-configurable platform.

CONCLUSIONS/RECOMMENDATIONS/SUMMARY

Novedu is a practice-based case study of how schools can move from exploratory use of generic AI tools toward institutionally embedded, teacher-configurable tutoring. Designing tutoring around educational workflows and reusable pedagogical modules appears better suited than generic chatbots to aligning AI support with instruction and giving teachers oversight, consistent with literature on pedagogical alignment and configurable AI tutoring (Sonkar et al., 2024; Song et al., 2025; Gabrovšek & Rihtaršič, 2025). The findings should inform responsible, pedagogically grounded adoption of generative AI in secondary education, and the cooperative-development model may transfer to other computer-science schools.

REFERENCES (OPTIONAL)

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KEYWORDS

generative AI in education; teacher-configurable AI tutors; pedagogical alignment; secondary education